INCH-POUND
MIL-F-24704/3(SH)
30 November 1988

# MILITARY SPECIFICATION SHEET

## FLANGES, FOUR BOLT SQUARE, BRAZED END FOR HYDRAULIC SYSTEMS

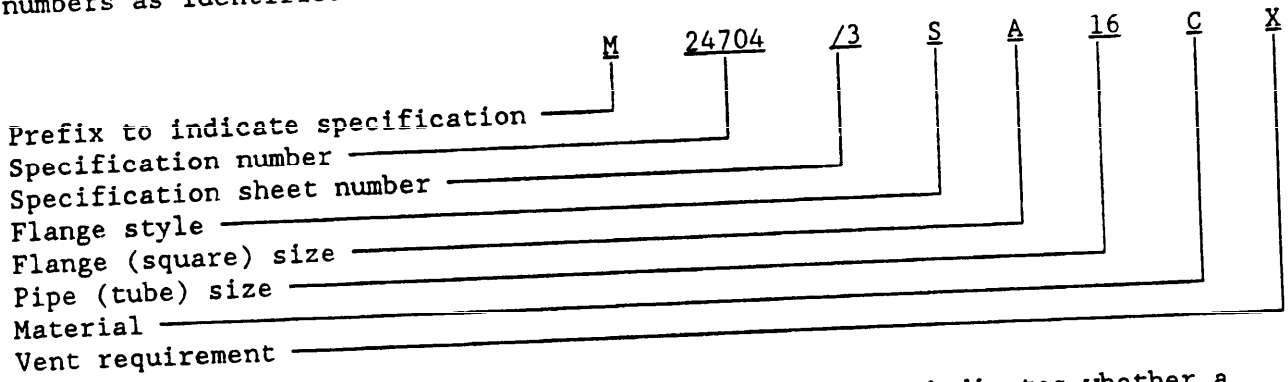
This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the flange described herein shall consist of this specification and the latest issue of MIL-F-24704(SH).

SCOPE

Scope. This specification sheet covers the requirements for four bolt square brazed end flanges for use in 3000 pounds per square inch ($lb/in^2$) hydraulic systems. These flanges are designed to utilize brazing rings conforming to MIL-F-20042 for pipe or Drawing 810-1385944 for tubing.

Classification. Brazed end flanges shall be designated by standard part numbers as identified below:




Flange style. The flange style designation letter indicates whether a straight or angle flange and whether for tubing or nominal pipe size (nps) piping as indicated below:

S = Straight flange for nps piping (see figure 1)
L = Right angle flange for nps piping (see figure 2)
T = Straight flange for outside diameter (od) tubing (see figure 3)
R = Right angle flange for od tubing (see figure 4)

FSC 4810
AMSC N/A
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:09Z
Check the source to verify that this is the current version before use.

Flange (square) size. The square size of the flange shall be designated by a letter in accordance with the following list (see figures 1, 2, 3 and 4 for listing of compatible flange and pipe (tube) sizes):

A = 1.50 inches square
B = 1.75
C = 2.00
D = 2.25
E = 2.75
F = 3.50
G = 4.25

Pipe (tube) size. The pipe or tubing size shall be designated as a two digit number in accordance with the following list:

| Size | Pipe (nps) | Tube (od) |
|---|---|---|
| 01 | 1/8 (0.405 od) | --- |
| 02 | 1/4 ( .540 od) | 0.250 |
| 03 | 3/8 ( .675 od) | .375 |
| 04 | 1/2 ( .840 od) | .500 |
| 06 | 3/4 (1.050 od) | .750 |
| 08 | 1 (1.315 od) | 1.000 |
| 10 | 1-1/4 (1.660 od) | 1.250 |
| 12 | 1-1/2 (1.900 od) | 1.500 |
| 16 | 2 (2.375 od) | 2.000 |

Material. The flange material shall be designated by a letter as indicated below:

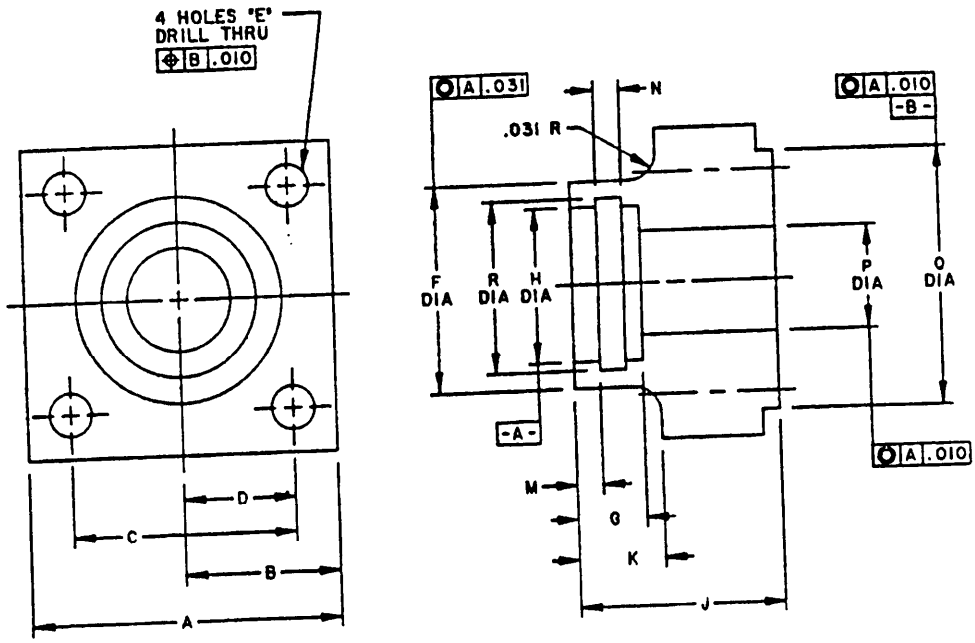
C = Corrosion resistant steel (CRES), class 304 or 316 in accordance with ASTM A 276, ASTM A 473 or QQ-S-763
L = Corrosion resistant steel (CRES), class 304L or 316L in accordance with ASTM A 276, ASTM A 473 or QQ-S-763

Vent requirement. The vent requirement shall be designated by a letter, X, Y or Z as indicated on figures 5, 6, and 7. For flanges without a vent the letter W shall be used. Machining for the vent port shall be as shown on the applicable figure.

Ultrasonic inspection. Flanges for pipe or tube, sizes 06 and larger, shall be suitable for ultrasonic inspection. The inspection area, which is the outer surface of the fitting socket, shall meet the 125 roughness height rating (RHR) or smoother finish requirement specified in the general specification. The inspection area shall extend from the face of the hub to a minimum distance equal to the socket depth (dimension G on figures 1 through 4). The corners of the socket hub face shall be squared rather than rounded as specified in the general specification. This inspection area shall be protected from damage during shipping by either a protective cap in accordance with MIL-C-5501 or hot dip strippable compound in accordance with MIL-P-23242.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:09Z
Check the source to verify that this is the current version before use.




UNLESS OTHERWISE SPECIFIED
ALL TOLERANCES SHALL BE +/- 0.030

PART NUMBER ---- M 24704/3 S# ** xx
# ** --- SELECTED FROM BELOW

| # | ** | FLANGE SIZE | PIPE SIZE N.P.S. | A SQ | B | C SQ | D | E | F +.015 -.000 | G +.015 -.000 | H +.003 -.000 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| B | 02 | 1.750 | 0.250 | 1.750 | 0.875 | 1.125 | 0.563 | 0.344 | 0.766 | 0.375 | 0.540 |
| C | 02 | 2.000 | 0.250 | 2.000 | 1.000 | 1.313 | 0.656 | 0.406 | 1.172 | 0.375 | 0.540 |
| C | 04 | 2.000 | 0.500 | 2.000 | 1.000 | 1.313 | 0.656 | 0.406 | 1.172 | 0.562 | 0.840 |
| D | 02 | 2.250 | 0.250 | 2.250 | 1.125 | 1.500 | 0.750 | 0.469 | 1.406 | 0.375 | 0.540 |
| D | 04 | 2.250 | 0.500 | 2.250 | 1.125 | 1.500 | 0.750 | 0.469 | 1.406 | 0.562 | 0.840 |
| D | 06 | 2.250 | 0.750 | 2.250 | 1.125 | 1.500 | 0.750 | 0.469 | 1.406 | 0.688 | 1.050 |
| E | 02 | 2.750 | 0.250 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 1.844 | 0.375 | 0.540 |
| E | 04 | 2.750 | 0.500 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 1.844 | 0.562 | 0.840 |
| E | 06 | 2.750 | 0.750 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 1.844 | 0.688 | 1.050 |
| E | 08 | 2.750 | 1.000 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 1.844 | 0.812 | 1.315 |
| F | 02 | 3.500 | 0.250 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.672 | 0.375 | 0.540 |
| F | 04 | 3.500 | 0.500 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.672 | 0.562 | 0.840 |
| F | 06 | 3.500 | 0.750 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.672 | 0.688 | 1.050 |
| F | 08 | 3.500 | 1.000 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.672 | 0.812 | 1.315 |
| F | 10 | 3.500 | 1.250 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.672 | 0.938 | 1.660 |
| F | 12 | 3.500 | 1.500 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.672 | 1.062 | 1.900 |

SH 13203063

FIGURE 1. Straight flange - pipe.



Source: https://assist.dla.mil -- Downloaded: 20260-20T01:09Z
Check the source to verify that this is the current version before use.

UNLESS OTHERWISE SPECIFIED
ALL TOLERANCES SHALL BE +/- 0.030

PART NUMBER ---- M 24704/3 S# ** xx
# ** --- SELECTED FROM BELOW

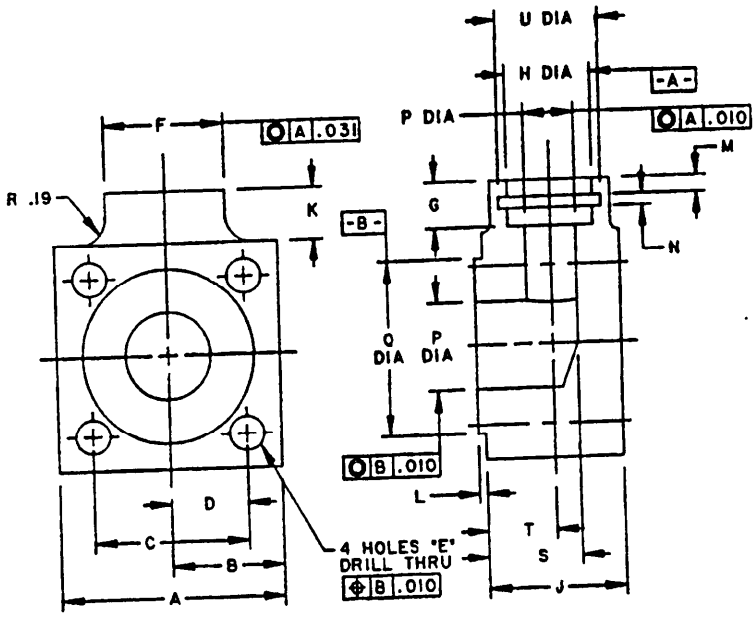
| # | ** | FLANGE SIZE | PIPE SIZE N.P.S. | J +.062 -.000 | K +.015 -.000 | L +.002 -.000 | M | N +.010 -.000 | P | Q +.000 -.002 | R +.020 -.000 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| B | 02 | 1.750 | 0.250 | 1.063 | 0.625 | 0.062 | 0.141 | 0.063 | 0.360 | 0.688 | 0.614 |
| C | 02 | 2.000 | 0.250 | 1.375 | 0.813 | 0.062 | 0.141 | 0.063 | 0.360 | 0.999 | 0.614 |
| C | 04 | 2.000 | 0.500 | 1.375 | 0.813 | 0.062 | 0.188 | 0.083 | 0.590 | 0.999 | 0.929 |
| D | 02 | 2.250 | 0.250 | 1.625 | 0.938 | 0.062 | 0.141 | 0.063 | 0.360 | 1.187 | 0.614 |
| D | 04 | 2.250 | 0.500 | 1.625 | 0.938 | 0.062 | 0.188 | 0.083 | 0.590 | 1.187 | 0.929 |
| D | 06 | 2.250 | 0.750 | 1.625 | 0.938 | 0.062 | 0.312 | 0.063 | 0.740 | 1.187 | 1.159 |
| E | 02 | 2.750 | 0.250 | 1.875 | 1.063 | 0.125 | 0.141 | 0.063 | 0.360 | 1.437 | 0.614 |
| E | 04 | 2.750 | 0.500 | 1.875 | 1.063 | 0.125 | 0.188 | 0.083 | 0.590 | 1.437 | 0.929 |
| E | 06 | 2.750 | 0.750 | 1.875 | 1.063 | 0.125 | 0.312 | 0.083 | 0.740 | 1.437 | 1.159 |
| E | 08 | 2.750 | 1.000 | 1.875 | 1.063 | 0.125 | 0.406 | 0.083 | 0.930 | 1.437 | 1.424 |
| F | 02 | 3.500 | 0.250 | 2.438 | 1.313 | 0.125 | 0.141 | 0.063 | 0.360 | 2.124 | 0.614 |
| F | 04 | 3.500 | 0.500 | 2.438 | 1.313 | 0.125 | 0.188 | 0.083 | 0.590 | 2.124 | 0.929 |
| F | 06 | 3.500 | 0.750 | 2.438 | 1.313 | 0.125 | 0.312 | 0.083 | 0.740 | 2.124 | 1.159 |
| F | 08 | 3.500 | 1.000 | 2.438 | 1.313 | 0.125 | 0.406 | 0.083 | 0.930 | 2.124 | 1.424 |
| F | 10 | 3.500 | 1.250 | 2.438 | 1.313 | 0.125 | 0.438 | 0.130 | 1.170 | 2.124 | 1.769 |
| F | 12 | 3.500 | 1.500 | 2.438 | 1.313 | 0.125 | 0.500 | 0.192 | 1.350 | 2.124 | 2.011 |

SH 13203063

FIGURE 1. Straight flange - pipe. - Continued



Source: https://assist.dla.mil -- Downloaded: 20261-20T01:09Z
Check the source to verify that this is the current version before use.




UNLESS OTHERWISE SPECIFIED
ALL TOLERANCES SHALL BE +/- 0.030

PART NUMBER ---- M 24704/3 L# ** xx
# ** --- SELECTED FROM BELOW

| # | ** | FLANGE SIZE | PIPE SIZE N.P.S. | A SQ | B | C SQ | D | E | F +.015 -.000 | G +.015 -.000 | H +.003 -.000 | J +.062 -.000 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B | 02 | 1.750 | 0.250 | 1.750 | 0.875 | 1.125 | 0.563 | 0.344 | 0.766 | 0.375 | 0.540 | 1.000 |
| C | 02 | 2.000 | 0.250 | 2.000 | 1.000 | 1.313 | 0.656 | 0.406 | 1.172 | 0.375 | 0.540 | 1.188 |
| C | 03 | 2.000 | 0.500 | 2.000 | 1.000 | 1.313 | 0.656 | 0.406 | 1.172 | 0.563 | 0.840 | 1.188 |
| D | 02 | 2.250 | 0.250 | 2.250 | 1.125 | 1.500 | 0.750 | 0.469 | 1.438 | 0.375 | 0.540 | 1.500 |
| D | 04 | 2.250 | 0.500 | 2.250 | 1.125 | 1.500 | 0.750 | 0.469 | 1.438 | 0.563 | 0.840 | 1.500 |
| D | 06 | 2.250 | 0.750 | 2.250 | 1.125 | 1.500 | 0.750 | 0.469 | 1.438 | 0.688 | 1.050 | 1.500 |
| E | 02 | 2.750 | 0.250 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 1.844 | 0.375 | 0.540 | 2.000 |
| E | 04 | 2.750 | 0.500 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 1.844 | 0.563 | 0.840 | 2.000 |
| E | 06 | 2.750 | 0.750 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 1.844 | 0.688 | 1.050 | 2.000 |
| E | 08 | 2.750 | 1.000 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 1.844 | 0.813 | 1.315 | 2.000 |
| F | 02 | 3.500 | 0.250 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.672 | 0.375 | 0.540 | 3.000 |
| F | 04 | 3.500 | 0.500 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.672 | 0.563 | 0.840 | 3.000 |
| F | 06 | 3.500 | 0.750 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.672 | 0.688 | 1.050 | 3.000 |
| F | 08 | 3.500 | 1.000 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.672 | 0.813 | 1.315 | 3.000 |
| F | 12 | 3.500 | 1.500 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.672 | 1.063 | 1.660 | 3.000 |

SH 13203064

FIGURE 2. Right angle flange - pipe.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:09Z
Check the source to verify that this is the current version before use.

UNLESS OTHERWISE SPECIFIED
ALL TOLERANCES SHALL BE +/- 0.030

PART NUMBER ---- M 24704/3 L# ** xx
# ** --- SELECTED FROM BELOW

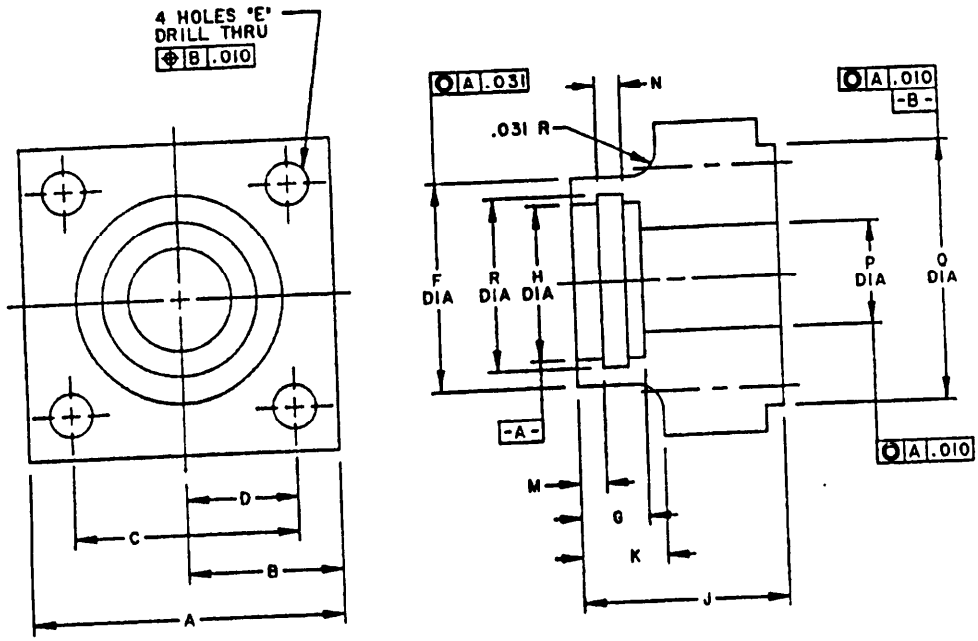
| # | ** | FLANGE SIZE | PIPE SIZE N.P.S. | K +.015 -.000 | L +.002 -.000 | M | N +.010 -.000 | P | Q +.000 -.002 | S | T | U |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B | 02 | 1.750 | 0.250 | 0.625 | 0.062 | 0.141 | 0.063 | 0.360 | 0.687 | 0.680 | 0.500 | 0.614 |
| C | 02 | 2.000 | 0.250 | 0.813 | 0.062 | 0.141 | 0.063 | 0.360 | 0.999 | 0.774 | 0.594 | 0.614 |
| C | 03 | 2.000 | 0.500 | 0.813 | 0.062 | 0.188 | 0.083 | 0.590 | 0.999 | 0.889 | 0.594 | 0.929 |
| D | 02 | 2.250 | 0.250 | 0.938 | 0.062 | 0.141 | 0.063 | 0.360 | 1.187 | 0.930 | 0.750 | 0.614 |
| D | 04 | 2.250 | 0.500 | 1.313 | 0.062 | 0.188 | 0.083 | 0.590 | 1.187 | 1.045 | 0.750 | 0.929 |
| D | 06 | 2.250 | 0.750 | 0.938 | 0.062 | 0.313 | 0.083 | 0.740 | 1.187 | 1.120 | 0.750 | 1.159 |
| E | 02 | 2.750 | 0.250 | 1.063 | 0.125 | 0.141 | 0.063 | 0.360 | 1.437 | 1.180 | 1.000 | 0.614 |
| E | 04 | 2.750 | 0.500 | 1.063 | 0.125 | 0.188 | 0.083 | 0.590 | 1.437 | 1.295 | 1.000 | 0.929 |
| E | 06 | 2.750 | 0.750 | 1.063 | 0.125 | 0.313 | 0.083 | 0.740 | 1.437 | 1.370 | 1.000 | 1.159 |
| E | 08 | 2.750 | 1.000 | 1.063 | 0.125 | 0.406 | 0.083 | 0.930 | 1.437 | 1.465 | 1.000 | 1.424 |
| F | 02 | 3.500 | 0.250 | 1.313 | 0.125 | 0.141 | 0.063 | 0.360 | 2.124 | 1.680 | 1.500 | 0.614 |
| F | 04 | 3.500 | 0.500 | 1.313 | 0.125 | 0.188 | 0.083 | 0.590 | 2.124 | 1.795 | 1.500 | 0.929 |
| F | 06 | 3.500 | 0.750 | 1.313 | 0.125 | 0.313 | 0.083 | 0.740 | 2.124 | 1.870 | 1.500 | 1.159 |
| F | 08 | 3.500 | 1.000 | 1.313 | 0.125 | 0.406 | 0.083 | 0.930 | 2.124 | 1.965 | 1.500 | 1.424 |
| F | 12 | 3.500 | 1.500 | 1.313 | 0.125 | 0.500 | 0.192 | 1.350 | 2.124 | 2.175 | 1.500 | 2.011 |

SH 13203064

FIGURE 2. Right angle flange - pipe. - Continued



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:09Z
Check the source to verify that this is the current version before use.




UNLESS OTHERWISE SPECIFIED
ALL TOLERANCES SHALL BE +/- 0.030

PART NUMBER ---- M 24704/3 T# ** xx
# ** --- SELECTED FROM BELOW

| # ** | FLANGE SIZE | PIPE SIZE N.P.S. | A SQ | B | C SQ | D | E | F +.015 -.000 | G +.015 -.000 | H +.000 -.003 |
|---|---|---|---|---|---|---|---|---|---|---|
| B 02 | 1.750 | 0.250 | 1.750 | 0.875 | 1.125 | 0.562 | 0.344 | 0.766 | 0.469 | 0.253 |
| B 04 | 1.750 | 0.500 | 1.750 | 0.875 | 1.125 | 0.562 | 0.344 | 0.766 | 0.469 | 0.503 |
| C 04 | 2.000 | 0.500 | 2.000 | 1.000 | 1.313 | 0.656 | 0.406 | 1.172 | 0.469 | 0.503 |
| C 06 | 2.000 | 0.750 | 2.000 | 1.000 | 1.313 | 0.656 | 0.406 | 1.172 | 0.562 | 0.753 |
| D 04 | 2.250 | 0.500 | 2.250 | 1.125 | 1.500 | 0.750 | 0.469 | 1.438 | 0.469 | 0.503 |
| D 06 | 2.250 | 0.750 | 2.250 | 1.125 | 1.500 | 0.750 | 0.469 | 1.438 | 0.562 | 0.753 |
| D 08 | 2.250 | 1.000 | 2.250 | 1.125 | 1.500 | 0.750 | 0.469 | 1.438 | 0.688 | 1.003 |
| E 04 | 2.750 | 0.500 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 2.188 | 0.469 | 0.503 |
| E 06 | 2.750 | 0.750 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 2.188 | 0.562 | 0.753 |
| E 08 | 2.750 | 1.000 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 2.188 | 0.688 | 1.003 |
| E 12 | 2.750 | 1.500 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 2.188 | 0.938 | 1.503 |
| F 04 | 3.500 | 0.500 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.812 | 0.469 | 0.503 |
| F 06 | 3.500 | 0.750 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.812 | 0.562 | 0.753 |
| F 08 | 3.500 | 1.000 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.812 | 0.688 | 1.003 |
| F 16 | 3.500 | 1.250 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.812 | 1.063 | 2.005 |

SH 13203065

FIGURE 3. Straight flange - tube.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:09Z
Check the source to verify that this is the current version before use.

UNLESS OTHERWISE SPECIFIED
ALL TOLERANCES SHALL BE +/- 0.030

PART NUMBER ---- M 24704/3 T# ** xx
# ** --- SELECTED FROM BELOW

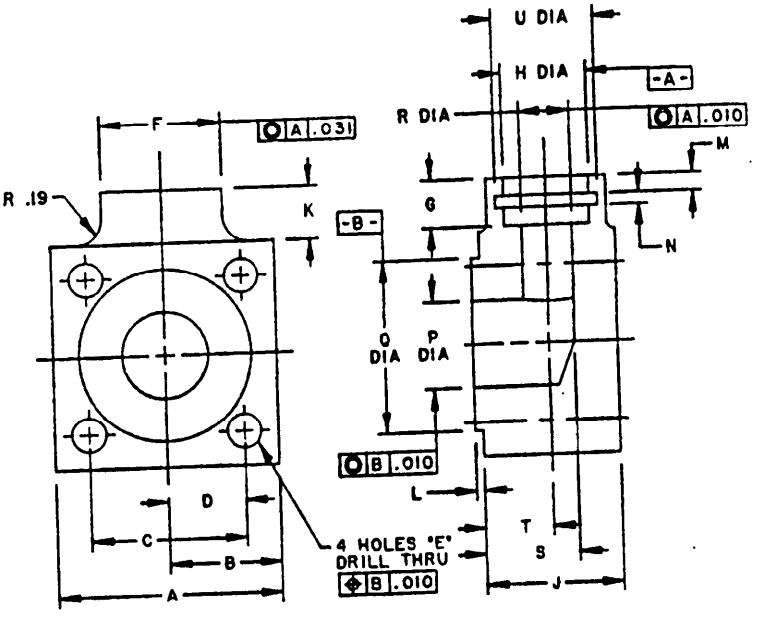
| # | ** | FLANGE SIZE | PIPE SIZE N.P.S. | J +.062 -.000 | K +.015 -.000 | L +.002 -.000 | M | N +.010 -.000 | P | Q +.000 -.002 | R +.020 -.000 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| B | 02 | 1.750 | 0.250 | 1.062 | 0.625 | 0.062 | 0.190 | 0.052 | 0.093 | 0.687 | 0.309 |
| B | 04 | 1.750 | 0.500 | 1.062 | 0.625 | 0.062 | 0.190 | 0.083 | 0.350 | 0.687 | 0.589 |
| C | 04 | 2.000 | 0.500 | 1.375 | 0.812 | 0.062 | 0.190 | 0.083 | 0.350 | 0.999 | 0.589 |
| C | 06 | 2.000 | 0.750 | 1.375 | 0.812 | 0.062 | 0.185 | 0.192 | 0.520 | 0.999 | 0.859 |
| D | 04 | 2.250 | 0.500 | 1.625 | 0.938 | 0.062 | 0.190 | 0.083 | 0.350 | 1.187 | 0.589 |
| D | 06 | 2.250 | 0.750 | 1.625 | 0.938 | 0.062 | 0.185 | 0.192 | 0.520 | 1.187 | 0.859 |
| D | 08 | 2.250 | 1.000 | 1.625 | 0.938 | 0.062 | 0.313 | 0.192 | 0.720 | 1.187 | 1.109 |
| E | 04 | 2.750 | 0.500 | 1.875 | 1.062 | 0.125 | 0.190 | 0.083 | 0.350 | 1.437 | 0.589 |
| E | 06 | 2.750 | 0.750 | 1.875 | 1.062 | 0.125 | 0.185 | 0.192 | 0.520 | 1.437 | 0.859 |
| E | 08 | 2.750 | 1.000 | 1.875 | 1.062 | 0.125 | 0.313 | 0.192 | 0.720 | 1.437 | 1.109 |
| E | 12 | 2.750 | 1.500 | 1.875 | 1.062 | 0.125 | 0.438 | 0.255 | 0.125 | 1.437 | 1.669 |
| F | 04 | 3.500 | 0.500 | 2.438 | 1.312 | 0.125 | 0.190 | 0.083 | 0.350 | 2.124 | 0.589 |
| F | 06 | 3.500 | 0.750 | 2.438 | 1.312 | 0.125 | 0.185 | 0.192 | 0.520 | 2.124 | 0.859 |
| F | 08 | 3.500 | 1.000 | 2.438 | 1.312 | 0.125 | 0.313 | 0.192 | 0.720 | 2.124 | 1.109 |
| F | 16 | 3.500 | 2.000 | 2.438 | 1.312 | 0.125 | 0.500 | 0.255 | 1.410 | 2.124 | 2.169 |

SH 13203065

FIGURE 3. Straight flange - tube. - Continued



Source: https://assist.dla.mil -- Downloaded: 202610-20T01:09Z
Check the source to verify that this is the current version before use.




UNLESS OTHERWISE SPECIFIED
ALL TOLERANCES SHALL BE +/- 0.030

PART NUMBER ---- M 24704/3 R# ** xx
# ** --- SELECTED FROM BELOW

| # | ** | FLANGE SIZE | PIPE SIZE N.P.S. | A SQ | B | C SQ | D | E | F +.015 -.000 | G +.015 -.000 | H +.000 -.003 | J +.062 -.000 | K +.015 -.000 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B | 02 | 1.750 | 0.250 | 1.750 | 0.875 | 1.125 | 0.562 | 0.344 | 0.766 | 0.469 | 0.253 | 1.000 | 0.500 |
| B | 04 | 1.750 | 0.500 | 1.750 | 0.875 | 1.125 | 0.562 | 0.344 | 0.766 | 0.469 | 0.503 | 1.000 | 0.500 |
| C | 04 | 2.000 | 0.500 | 2.000 | 1.000 | 1.312 | 0.656 | 0.406 | 1.172 | 0.469 | 0.503 | 1.188 | 0.594 |
| C | 06 | 2.000 | 0.750 | 2.000 | 1.000 | 1.312 | 0.656 | 0.406 | 1.172 | 0.562 | 0.753 | 1.188 | 0.594 |
| D | 04 | 2.250 | 0.500 | 2.250 | 1.125 | 1.500 | 0.750 | 0.469 | 1.438 | 0.469 | 0.503 | 1.500 | 0.750 |
| D | 06 | 2.250 | 0.750 | 2.250 | 1.125 | 1.500 | 0.750 | 0.469 | 1.438 | 0.562 | 0.753 | 1.500 | 0.750 |
| D | 08 | 2.250 | 1.000 | 2.250 | 1.125 | 1.500 | 0.750 | 0.469 | 1.438 | 0.688 | 1.003 | 1.500 | 0.750 |
| E | 04 | 2.750 | 0.500 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 2.188 | 0.469 | 0.503 | 2.250 | 1.125 |
| E | 06 | 2.750 | 0.750 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 2.188 | 0.562 | 0.753 | 2.250 | 1.125 |
| E | 08 | 2.750 | 1.000 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 2.188 | 0.688 | 1.003 | 2.250 | 1.125 |
| E | 12 | 2.750 | 1.500 | 2.750 | 1.375 | 1.875 | 0.938 | 0.469 | 2.188 | 0.938 | 1.503 | 2.250 | 1.125 |
| F | 04 | 3.500 | 0.500 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.812 | 0.469 | 0.503 | 3.000 | 1.500 |
| F | 06 | 3.500 | 0.750 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.812 | 0.562 | 0.753 | 3.000 | 1.500 |
| F | 08 | 3.500 | 1.000 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.812 | 0.688 | 1.003 | 3.000 | 1.500 |
| F | 12 | 3.500 | 1.500 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.812 | 0.938 | 1.503 | 3.000 | 1.500 |
| F | 16 | 3.500 | 2.000 | 3.500 | 1.750 | 2.500 | 1.250 | 0.531 | 2.812 | 1.062 | 2.005 | 3.000 | 1.500 |

SH 13203066

FIGURE 4. Right angle flange - tube.



Source: https://assist.dla.mil -- Downloaded: 202610-20T01:09Z
Check the source to verify that this is the current version before use.

UNLESS OTHERWISE SPECIFIED
ALL TOLERANCES SHALL BE +/- 0.030

PART NUMBER ---- M 24704/3 R# ** xx
# ** --- SELECTED FROM BELOW

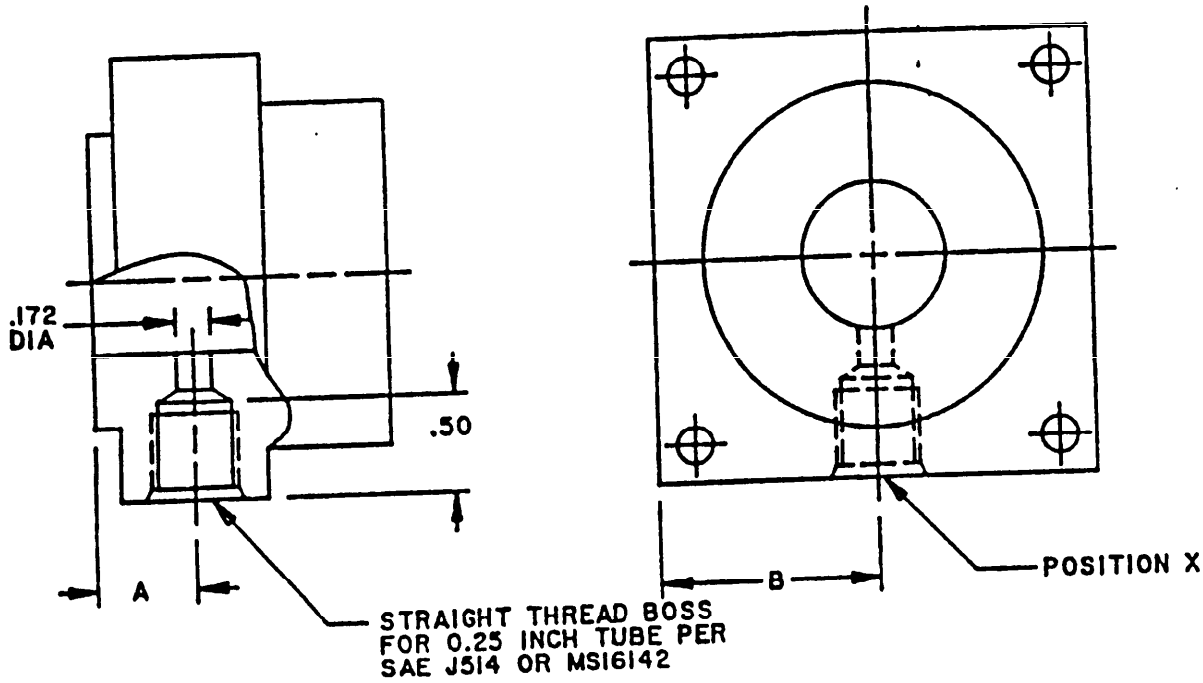
| # | ** | FLANGE SIZE | PIPE SIZE N.P.S. | L +.002 -.000 | M | N +.010 -.000 | P | Q +.000 -.002 | R | S | T | U +.020 -.000 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| B | 02 | 1.750 | 0.250 | 0.062 | 0.190 | 0.052 | 0.250 | 0.687 | 0.093 | 0.547 | 0.500 | 0.309 |
| B | 04 | 1.750 | 0.500 | 0.062 | 0.190 | 0.083 | 0.350 | 0.687 | 0.350 | 0.675 | 0.500 | 0.589 |
| C | 04 | 2.000 | 0.500 | 0.062 | 0.190 | 0.083 | 0.350 | 0.999 | 0.350 | 0.769 | 0.594 | 0.589 |
| C | 06 | 2.000 | 0.750 | 0.062 | 0.185 | 0.192 | 0.520 | 0.999 | 0.520 | 0.854 | 0.594 | 0.859 |
| D | 04 | 2.250 | 0.500 | 0.062 | 0.190 | 0.083 | 0.350 | 1.187 | 0.350 | 0.925 | 0.750 | 0.589 |
| D | 06 | 2.250 | 0.750 | 0.062 | 0.185 | 0.192 | 0.520 | 1.187 | 0.520 | 1.010 | 0.750 | 0.859 |
| D | 08 | 2.250 | 1.000 | 0.062 | 0.312 | 0.192 | 0.720 | 1.187 | 0.720 | 1.110 | 0.750 | 1.109 |
| E | 04 | 2.750 | 0.500 | 0.125 | 0.190 | 0.083 | 0.350 | 1.437 | 0.350 | 1.300 | 1.125 | 0.589 |
| E | 06 | 2.750 | 0.750 | 0.125 | 0.185 | 0.192 | 0.520 | 1.437 | 0.520 | 1.385 | 1.125 | 0.859 |
| E | 08 | 2.750 | 1.000 | 0.125 | 0.312 | 0.192 | 0.720 | 1.437 | 0.720 | 1.485 | 1.125 | 0.109 |
| E | 12 | 2.750 | 1.500 | 0.125 | 0.438 | 0.255 | 1.080 | 1.437 | 1.080 | 1.665 | 1.125 | 1.669 |
| F | 04 | 3.500 | 0.500 | 0.125 | 0.190 | 0.083 | 0.350 | 2.124 | 0.350 | 1.675 | 1.500 | 0.589 |
| F | 06 | 3.500 | 0.750 | 0.125 | 0.185 | 0.192 | 0.520 | 2.124 | 0.520 | 1.760 | 1.500 | 0.859 |
| F | 08 | 3.500 | 1.000 | 0.125 | 0.312 | 0.192 | 0.720 | 2.124 | 0.720 | 1.860 | 1.500 | 1.109 |
| F | 12 | 3.500 | 1.500 | 0.125 | 0.438 | 0.255 | 1.080 | 2.124 | 1.080 | 2.040 | 1.500 | 1.669 |
| F | 16 | 3.500 | 2.000 | 0.125 | 0.500 | 0.255 | 1.410 | 2.124 | 1.410 | 2.205 | 1.500 | 2.169 |

SH 13203066

FIGURE 4. Right angle flange - tube. - Continued



Source: https://assist.dla.mil -- Downloaded: 202610-20T01:09Z
Check the source to verify that this is the current version before use.




THE VENT PORT CAN BE LOCATED AT POSITION X FOR THE FOLLOWING FLANGES. INSERT THE LETTER X AT THE APPROPRIATE LOCATION IN THE FLANGE PART NUMBER. FOR FLANGES WITHOUT VENT REQUIREMENT, INSERT THE LETTER W.

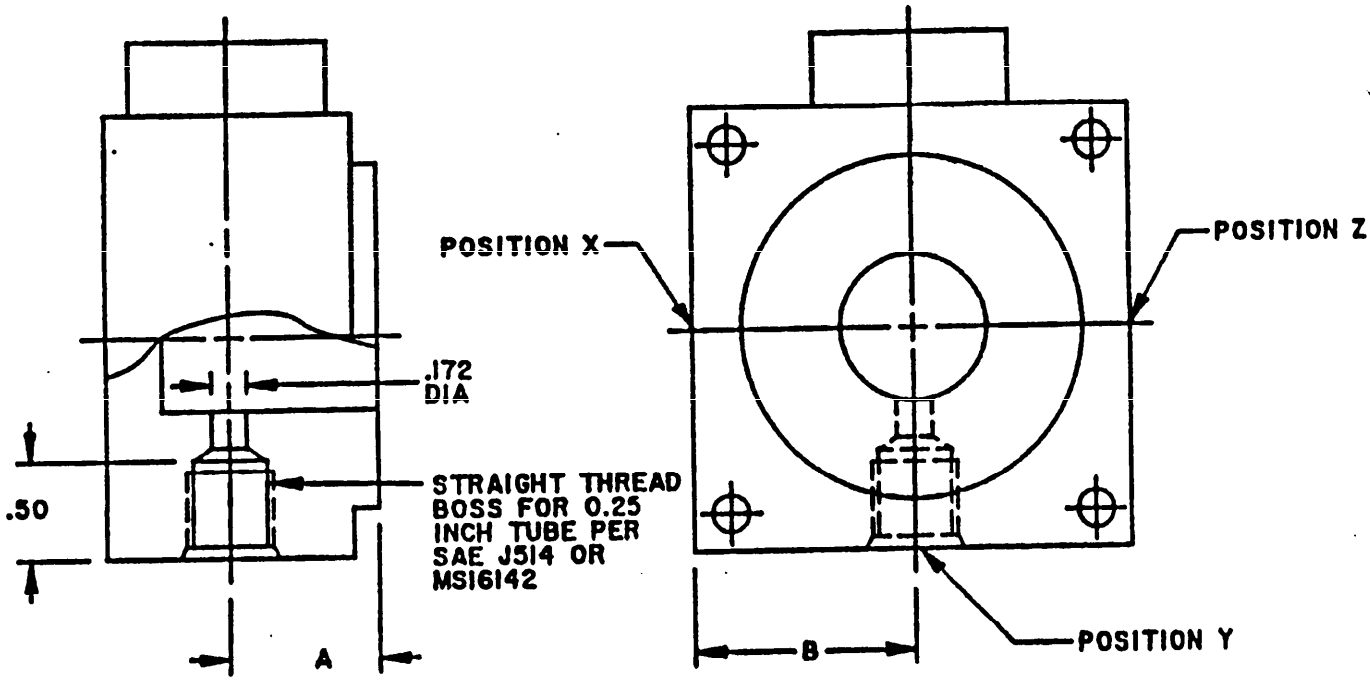
UNLESS OTHERWISE SPECIFIED
ALL TOLERANCES SHALL BE +/- 0.030

| # ** | FLANGE SIZE (INCH) | PIPE SIZE N.P.S. | A | B |
|---|---|---|---|---|
| F 02 | 3.500 | 0.250 | 0.660 | 1.750 |
| F 04 | 3.500 | 0.500 | 0.660 | 1.750 |
| F 06 | 3.500 | 0.750 | 0.660 | 1.750 |
| F 08 | 3.500 | 1.000 | 0.660 | 1.750 |
| F 10 | 3.500 | 1.250 | 0.660 | 1.750 |
| F 12 | 3.500 | 1.500 | 0.660 | 1.750 |

SH 13203067

FIGURE 5. Straight flange vent location - pipe.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:09Z
Check the source to verify that this is the current version before use.




THE VENT PORT CAN BE LOCATED AT POSITION X,Y, OR Z FOR THE FOLLOWING FLANGES. INSERT THE LETTER X,Y, OR Z AT THE APPROPRIATE LOCATION IN THE FLANGE PART NUMBER. FOR FLANGES WITHOUT VENT REQUIREMENT, INSERT THE LETTER W.

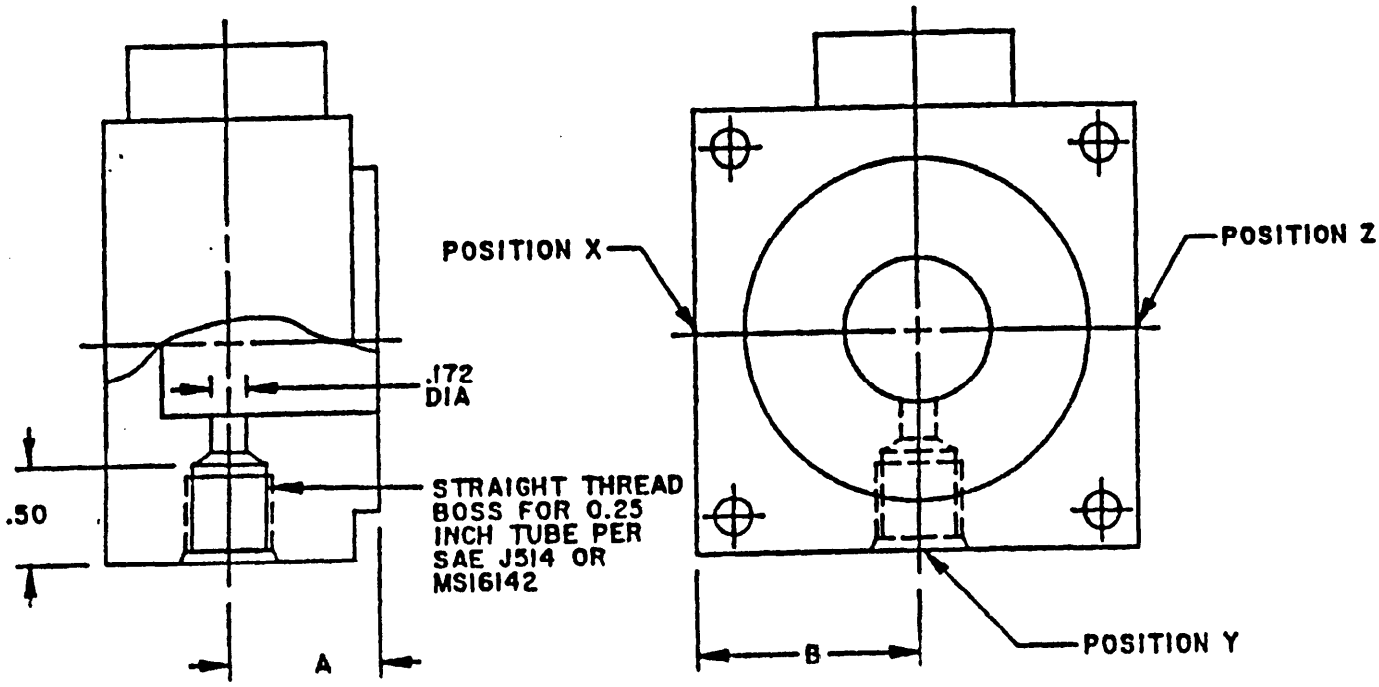
UNLESS OTHERWISE SPECIFIED
ALL TOLERANCES SHALL BE +/- 0.030

| # ** | FLANGE SIZE (INCH) | PIPE SIZE N.P.S. | A | B |
|---|---|---|---|---|
| D 02 | 2.250 | 0.250 | 0.812 | 1.125 |
| D 04 | 2.250 | 0.500 | 0.812 | 1.125 |
| D 06 | 2.250 | 0.750 | 0.812 | 1.125 |
| E 02 | 2.750 | 0.250 | 1.125 | 1.375 |
| E 04 | 2.750 | 0.500 | 1.125 | 1.375 |
| E 06 | 2.750 | 0.750 | 1.125 | 1.375 |
| E 08 | 2.750 | 1.000 | 1.125 | 1.375 |
| F 02 | 3.500 | 0.250 | 1.625 | 1.750 |
| F 04 | 3.500 | 0.500 | 1.625 | 1.750 |
| F 06 | 3.500 | 0.750 | 1.625 | 1.750 |
| F 08 | 3.500 | 1.000 | 1.625 | 1.750 |
| F 12 | 3.500 | 1.500 | 1.625 | 1.750 |

SH 13203068

FIGURE 6. Right angle flange vent location - pipe.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:09Z
Check the source to verify that this is the current version before use.




THE VENT PORT CAN BE LOCATED AT POSITION X,Y, OR Z FOR THE FOLLOWING FLANGES. INSERT THE LETTER X,Y, OR Z AT THE APPROPRIATE LOCATION IN THE FLANGE PART NUMBER. FOR FLANGES WITHOUT VENT REQUIREMENT, INSERT THE LETTER W.

UNLESS OTHERWISE SPECIFIED
ALL TOLERANCES SHALL BE +/- 0.030

| # ** | FLANGE SIZE (INCH) | TUBE SIZE | A | B |
|---|---|---|---|---|
| F 12 | 2.750 | 1.500 | 1.250 | 1.375 |
| F 12 | 3.500 | 1.500 | 1.625 | 1.750 |
| F 16 | 3.500 | 2.000 | 1.625 | 1.750 |

SH 13203069

FIGURE 7. Right angle flange vent location - tube.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:09Z
Check the source to verify that this is the current version before use.

Preparing activity:
Navy - SH
(Project 4810-N060-3)



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:09Z
Check the source to verify that this is the current version before use.